

This artifact is a C++ Hangman game I built a few years ago for CIS 125 at Berkshire Community College. For this round of improvements, I focused on how the program stores and retrieves words. I swapped out the old text file for a SQLite database. Now when the computer is the executioner the player gets to pick a difficulty and the program pulls a word from the database based on length. Easy is 3 to 5 letters, medium is 6 to 7, and hard is 8 or more. All the database logic lives in a new database manager class which slots right into the structure I already set up in the previous enhancement.

I kept coming back to this project because I wanted to make all my changes to the same program so I could see the improvements build on each other over time. Connecting a real database to a working program is something that comes up in software development so being able to show I can do that feels more valuable than a lot of other things I could have added. Replacing a text file with a structured SQLite database speaks to using real industry tools and making deliberate design choices which ties into a few of the course outcomes around building solutions that deliver value. The hardest part was just getting SQLite to cooperate with Xcode. Figuring out how library linking works and where the database file ends up at runtime took some trial and error. But working through that was probably the most useful part because it forced me to understand things.